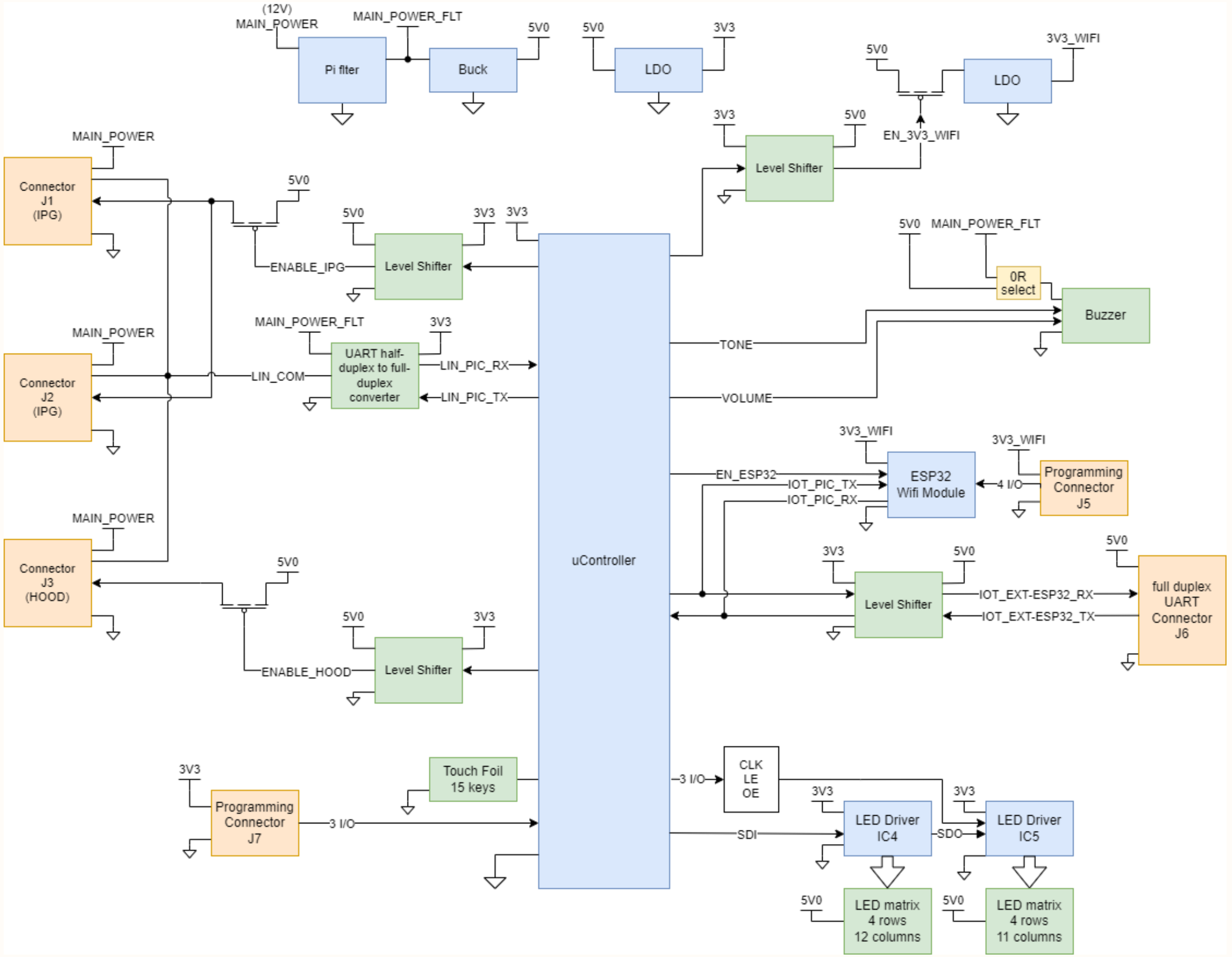
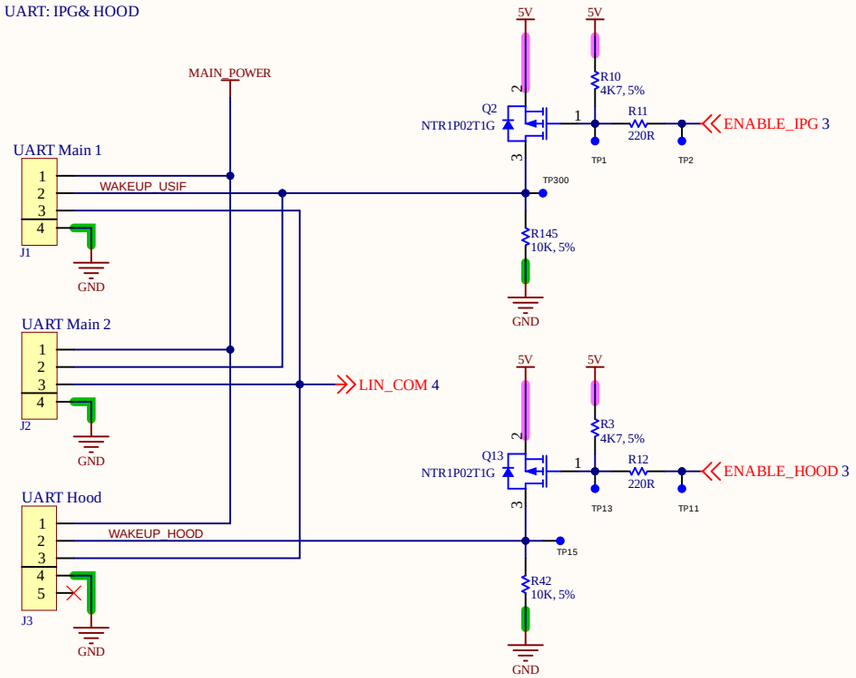
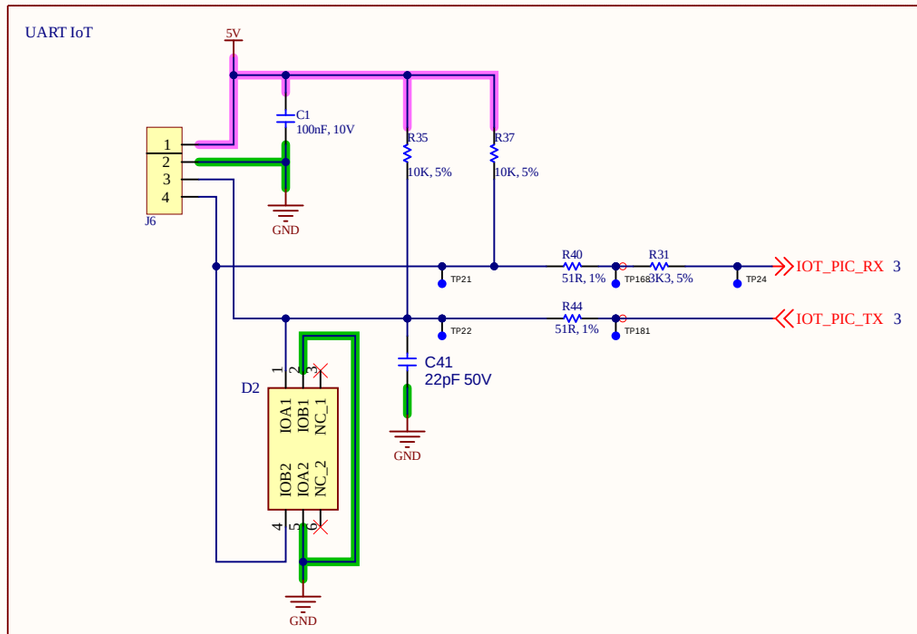

(12V)
MAIN_POWER
MAIN_POWER_FLT
Pi filter
Buck
5V0
5V0
LDO
3V3
5V0
LDO
3V3_WIFI
EN_3V3_WIFI
3V3
5V0
Level Shifter
MAIN_POWER
Connector
J1
(IPG)
5V0
ENABLE_IPG
5V0
3V3
3V3
Level Shifter
5V0
MAIN_POWER_FLT
0R
select
Buzzer
MAIN_POWER
Connector
J2
(IPG)
MAIN_POWER_FLT
3V3
UART half-
duplex to full-
duplex
converter
LIN_COM
LIN_PIC_RX
LIN_PIC_TX
TONE
VOLUME
3V3_WIFI
3V3_WIFI
EN_ESP32
IOT_PIC_TX
IOT_PIC_RX
ESP32
Wifi Module
4 I/O
Programming
Connector
J5
MAIN_POWER
Connector
J3
(HOOD)
5V0
3V3
5V0
Level Shifter
IOT_EXT-ESP32_RX
IOT_EXT-ESP32_TX
5V0
full duplex
UART
Connector
J6
5V0
3V3
Level Shifter
ENABLE_HOOD
uController
3V3
Programming
Connector
J7
3 I/O
Touch Foil
15 keys
3 I/O
CLK
LE
OE
SDI
3V3
LED Driver
IC4
SDO
3V3
LED Driver
IC5
5V0
LED matrix
4 rows
12 columns
5V0
LED matrix
4 rows
11 columns

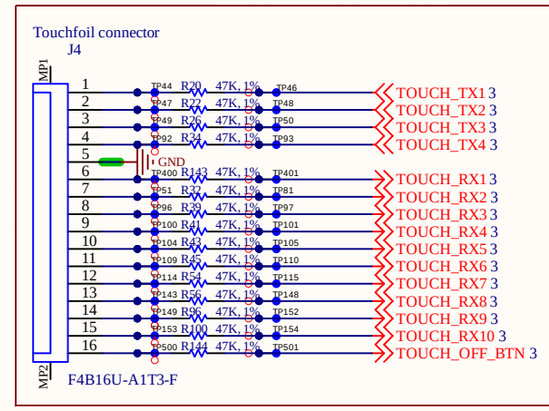
UART: IPG& HOOD



UART IoT



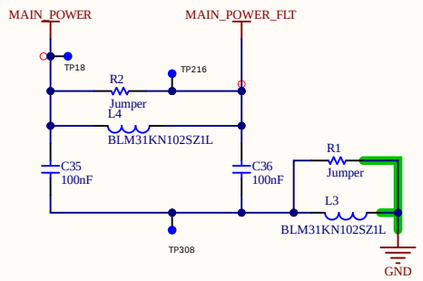
Touchfoil connector



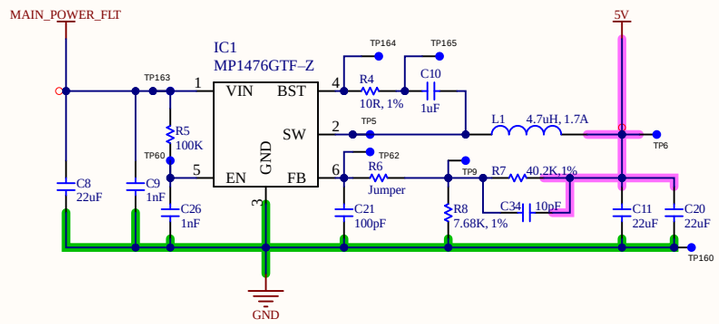


TP157 GND
TP210 FVT_PIC_RX_3V3 3
TP211 FVT_PIC_TX_3V3 3

Main Power Filter



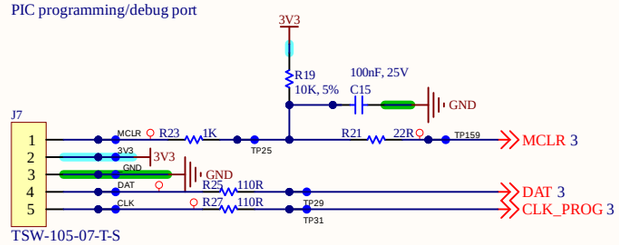
12V->5V Buck regulator



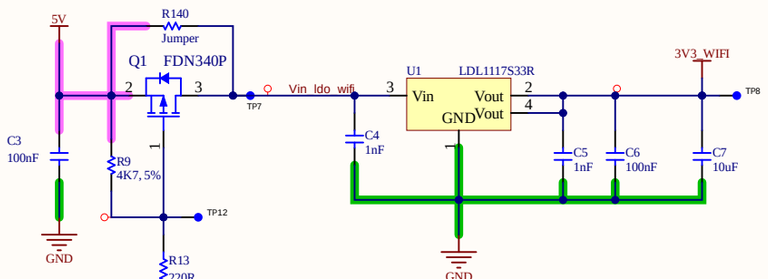
ESP32 programming port



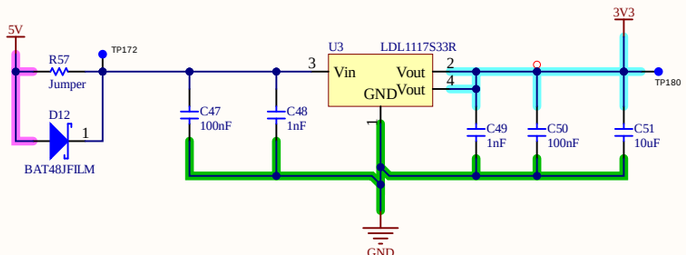
PIC programming/debug port



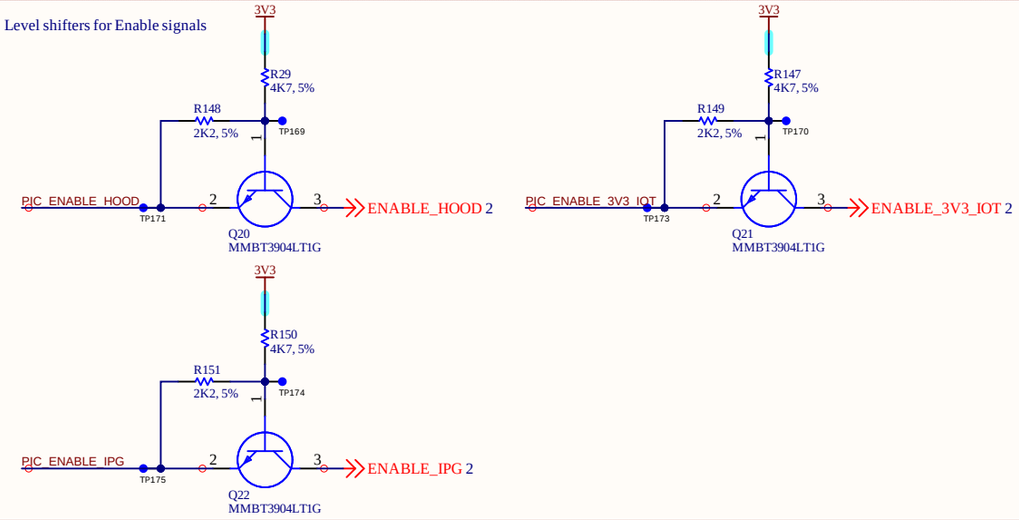
5V->3V3 LDO (WiFi)



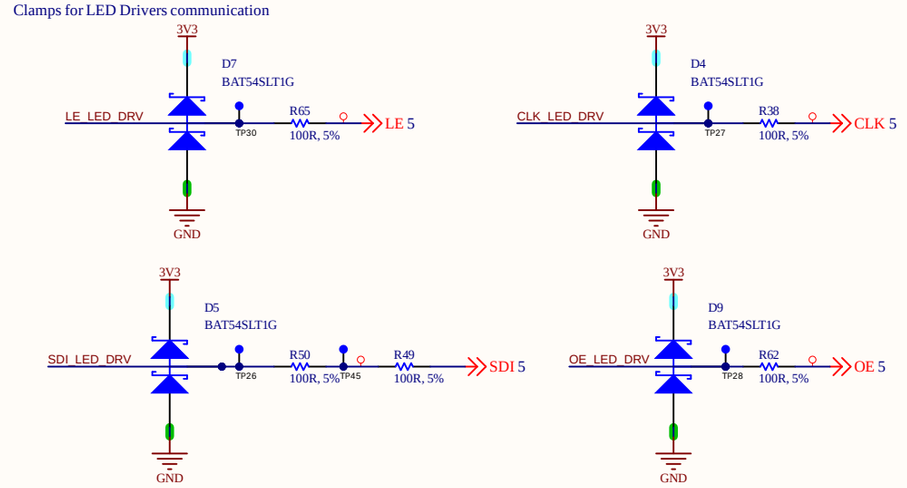
5V->3V3 LDO (uC+LED drivers)



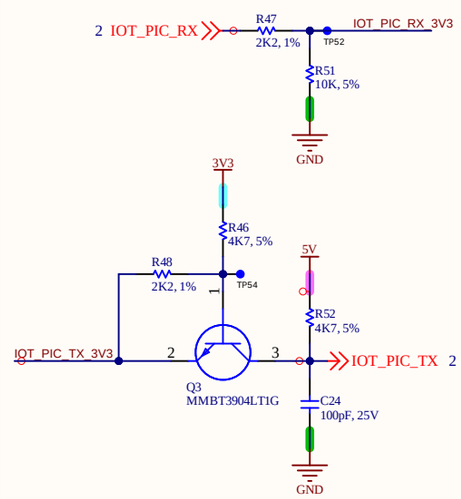
Level shifters for Enable signals



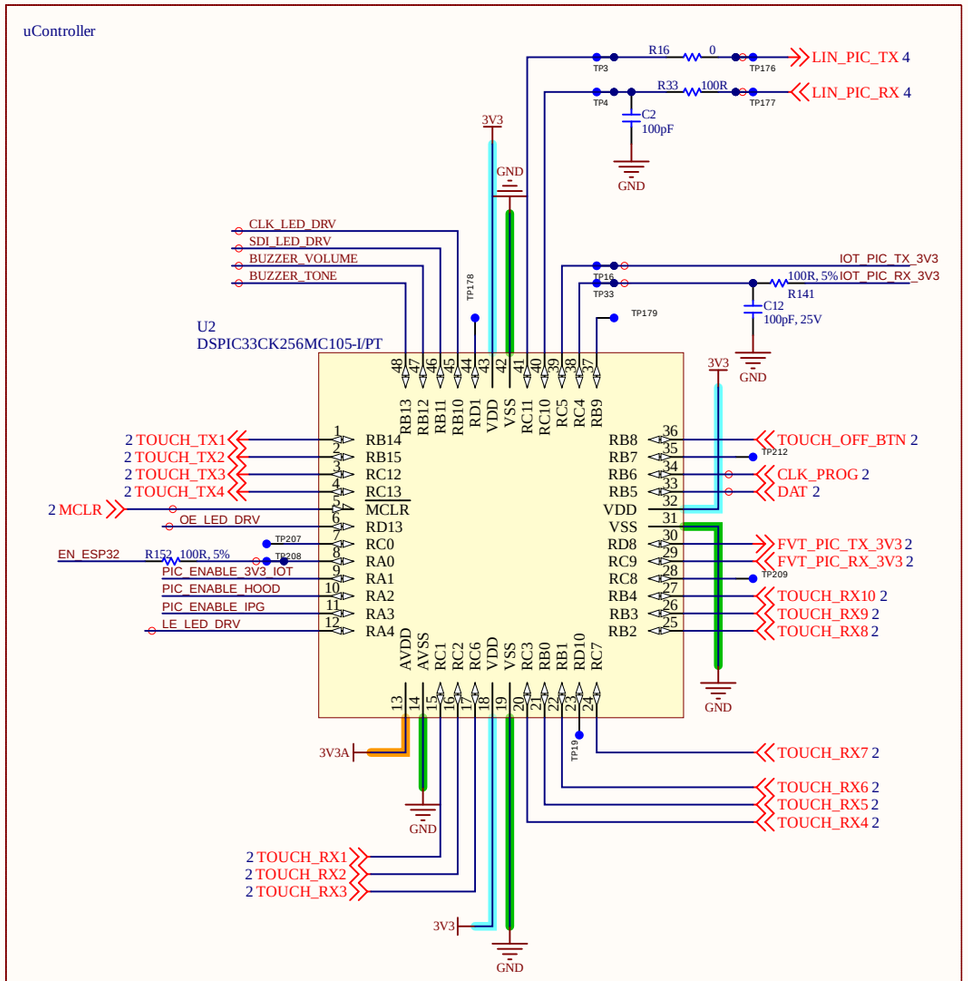
Clamps for LED Drivers communication

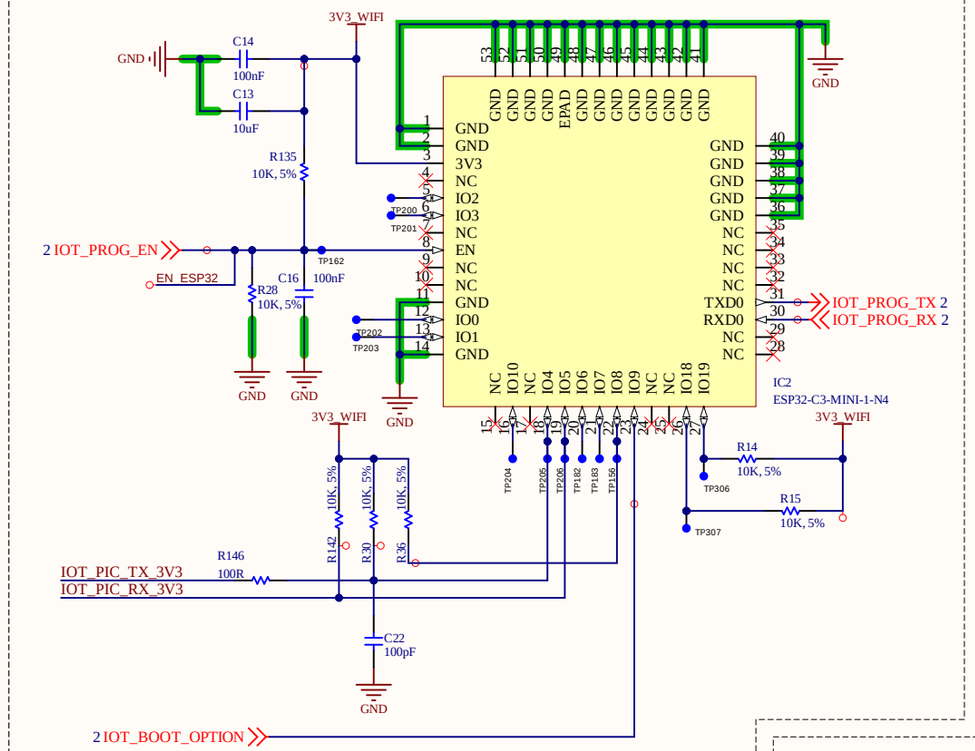


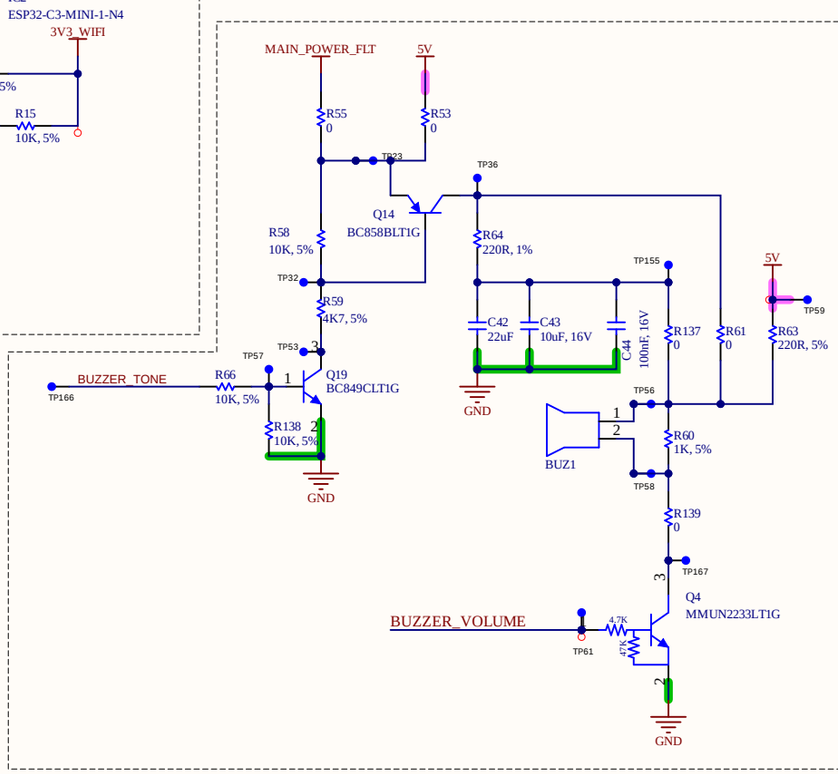
UART IOT - Level shifters to external module



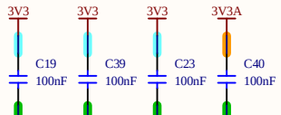
uController

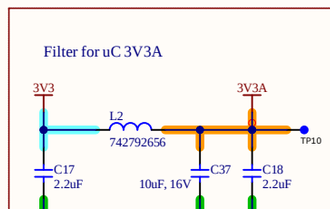


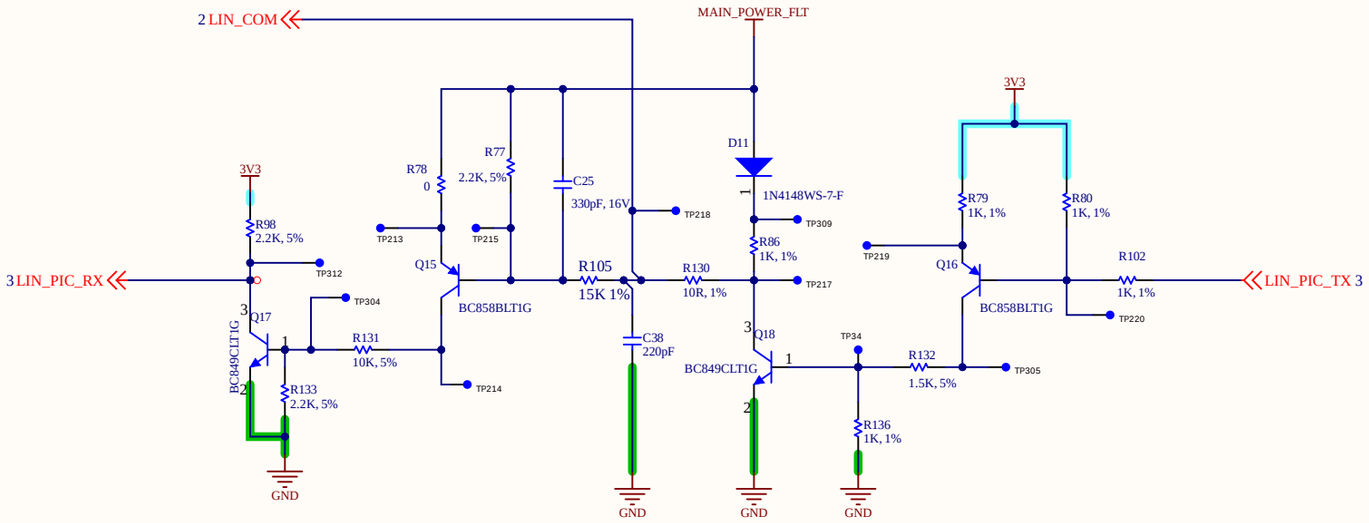


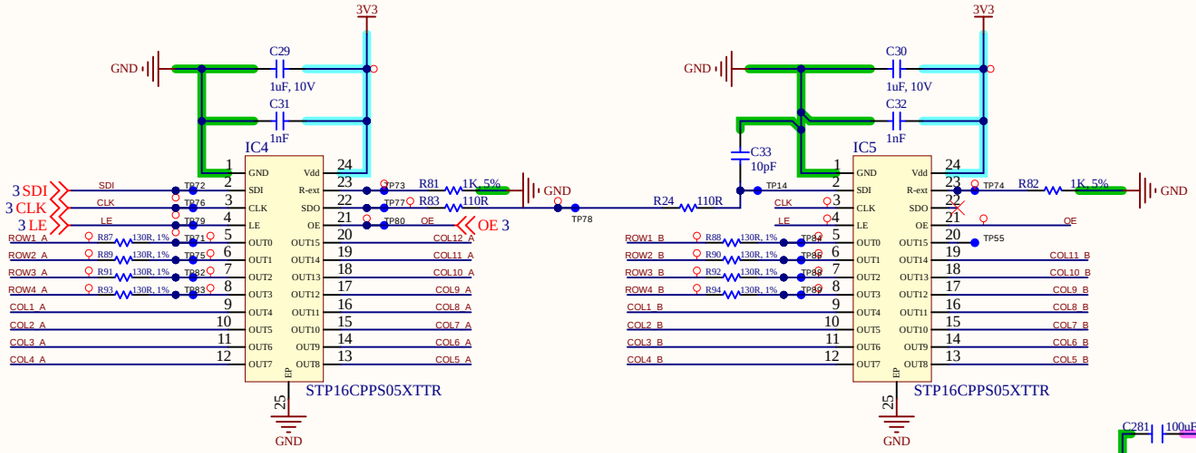


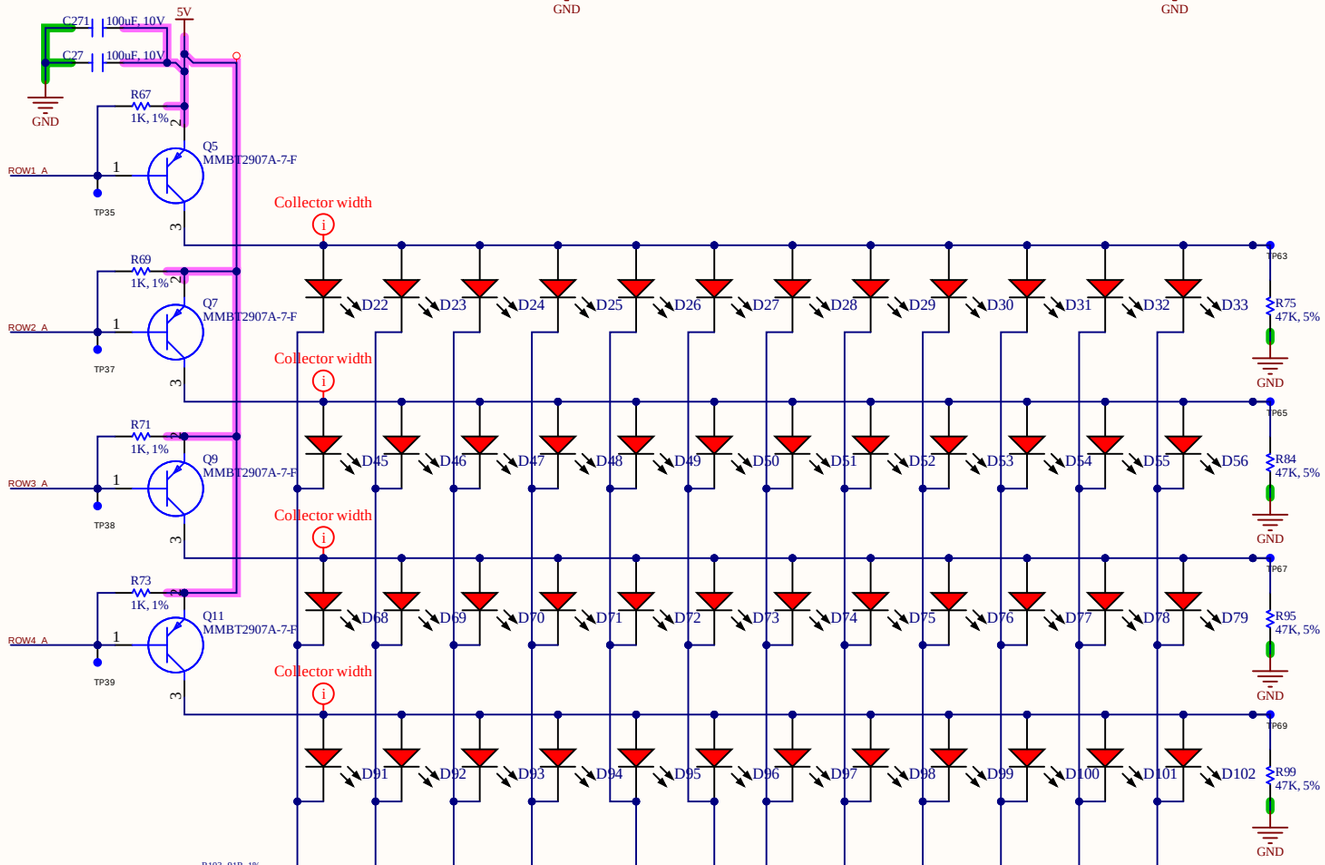
Keep C19 C39 C23 C40 close to uC pins

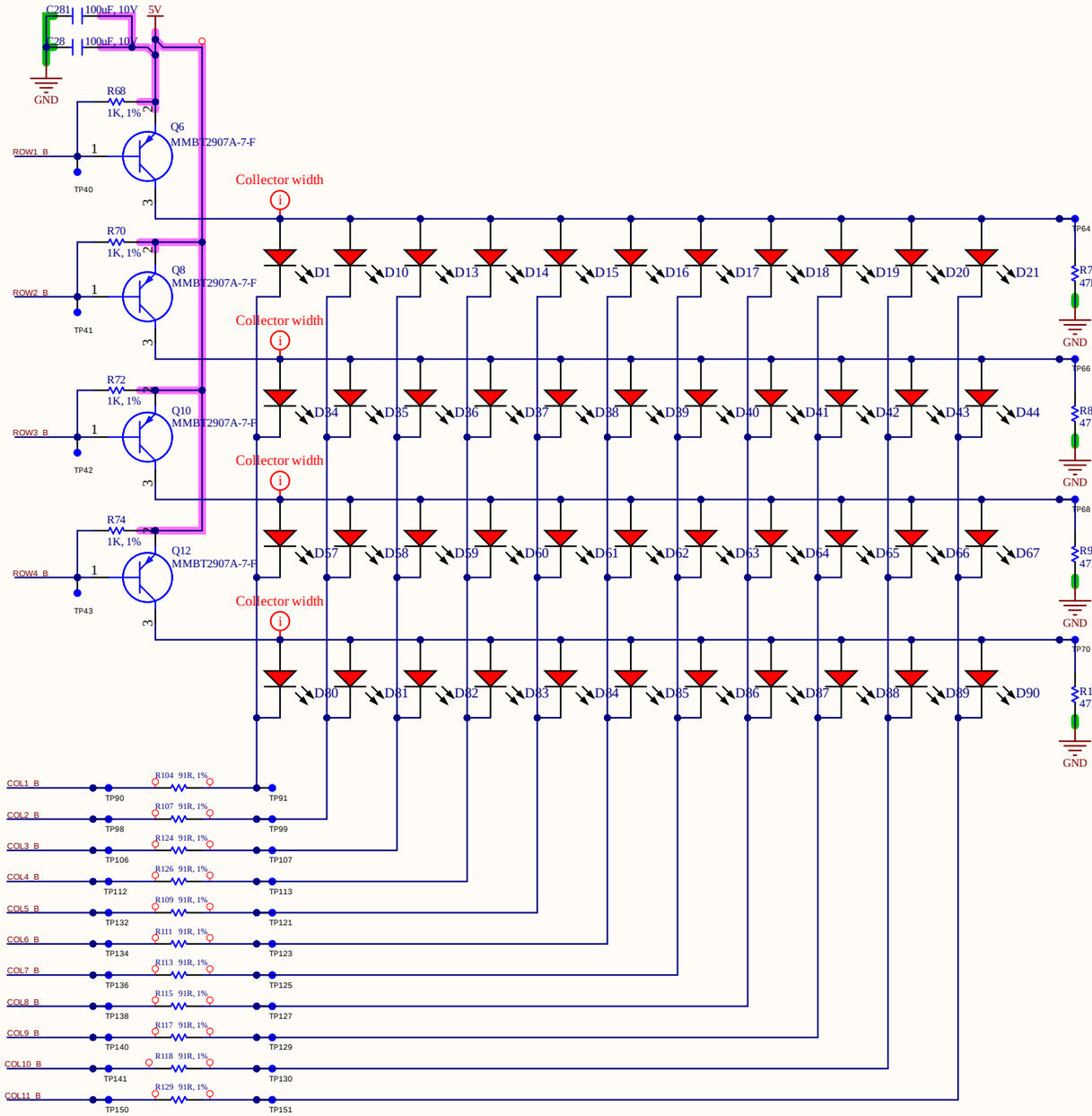


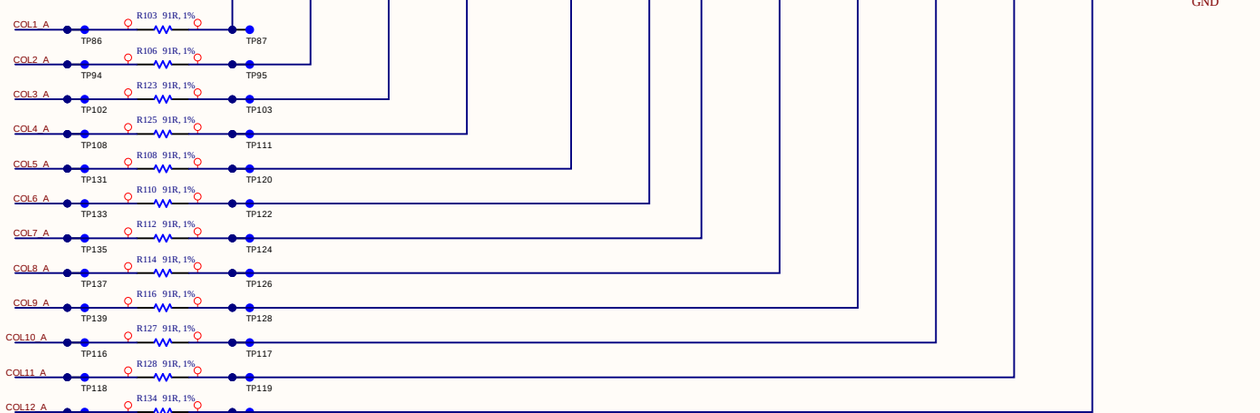



2 LIN_COM
MAIN_POWER_FLT
3V3
R78
0
R77
2.2K, 5%
C25
330pF, 16V
TP218
D11
1N4148WS-7-F
TP309
R86
1K, 1%
TP213
TP215
TP219
R79
1K, 1%
R80
1K, 1%
3V3
R98
2.2K, 5%
TP312
Q15
R105
15K 1%
R130
10R, 1%
TP217
Q16
R102
1K, 1%
TP220
3 LIN_PIC_RX
LIN_PIC_TX 3
BC858BLT1G
BC858BLT1G
TP304
Q17
BC849CLT1G
R131
10K, 5%
TP214
C38
220pF
Q18
BC849CLT1G
TP34
R132
1.5K, 5%
TP305
R133
2.2K, 5%
R136
1K, 1%
GND
GND
GND
GND


3V3
GND
C29
1uF, 10V
C31
1nF
IC4
STP16CPPS05XTTR
3 SDI
3 CLK
3 LE
OE 3
R81 1K, 5%
R83 110R
TP78
C33
10pF
R24 110R
IC5
R82 1K, 5%
C30
C32
TP55
ROW1_A
ROW2_A
ROW3_A
ROW4_A
COL1_A
COL2_A
COL3_A
COL4_A
ROW1_B
ROW2_B
ROW3_B
ROW4_B
COL1_B
COL2_B
COL3_B
COL4_B


C271 100uF, 10V
C27 100uF, 10V
5V
GND
R67 1K, 1%
Q5 MMBT2907A-7-F
ROW1_A
TP35
R69 1K, 1%
Q7 MMBT2907A-7-F
ROW2_A
TP37
R71 1K, 1%
Q9 MMBT2907A-7-F
ROW3_A
TP38
R73 1K, 1%
Q11 MMBT2907A-7-F
ROW4_A
TP39
Collector width
D22 D23 D24 D25 D26 D27 D28 D29 D30 D31 D32 D33
D45 D46 D47 D48 D49 D50 D51 D52 D53 D54 D55 D56
D68 D69 D70 D71 D72 D73 D74 D75 D76 D77 D78 D79
D91 D92 D93 D94 D95 D96 D97 D98 D99 D100 D101 D102
TP63
R75 47K, 5%
TP65
R84 47K, 5%
TP67
R95 47K, 5%
TP69
R99 47K, 5%


C281 100uF, 10V
C28 100uF, 10V
5V
GND
R68 1K, 1%
Q6 MMBT2907A-7-F
ROW1_B
TP40
R70 1K, 1%
Q8 MMBT2907A-7-F
ROW2_B
TP41
R72 1K, 1%
Q10 MMBT2907A-7-F
ROW3_B
TP42
R74 1K, 1%
Q12 MMBT2907A-7-F
ROW4_B
TP43
Collector width
D1 D10 D13 D14 D15 D16 D17 D18 D19 D20 D21
D34 D35 D36 D37 D38 D39 D40 D41 D42 D43 D44
D57 D58 D59 D60 D61 D62 D63 D64 D65 D66 D67
D80 D81 D82 D83 D84 D85 D86 D87 D88 D89 D90
TP64
TP66
TP68
TP70
COL1_B
COL2_B
COL3_B
COL4_B
COL5_B
COL6_B
COL7_B
COL8_B
COL9_B
COL10_B
COL11_B
TP90 TP91
TP98 TP99
TP106 TP107
TP112 TP113
TP132 TP121
TP134 TP123
TP136 TP125
TP138 TP127
TP140 TP129
TP141 TP130
TP150 TP151


COL1_A R103 91R, 1% TP86 TP87
COL2_A R106 91R, 1% TP94 TP95
COL3_A R123 91R, 1% TP102 TP103
COL4_A R125 91R, 1% TP108 TP111
COL5_A R108 91R, 1% TP131 TP120
COL6_A R110 91R, 1% TP133 TP122
COL7_A R112 91R, 1% TP135 TP124
COL8_A R114 91R, 1% TP137 TP126
COL9_A R116 91R, 1% TP139 TP128
COL10_A R127 91R, 1% TP116 TP117
COL11_A R128 91R, 1% TP118 TP119
COL12_A R134 91R, 1%